Pandas Cheatsheet — Essential Functions

Data Loading & Inspection	
pd.read_csv("file.csv")	Load CSV
pd.read_json("data.json")	Load JSON
pd.read_excel("file.xlsx")	Load Excel
df.info()	Schema + nulls
df.head(10)	Preview rows
df.tail(5)	Last rows
df.sample(5)	Random sample
df.shape	Rows/columns
df.dtypes	Column types
df.columns	Column names
df.describe()	Summary stats

Selecting & Filtering	
df['col']	Select column
df[['a','b']]	Select multiple columns
df.iloc[0:5]	Select by position
df.loc[df['x']>0]	Select by label/condition
df[df['age'] > 30]	Boolean filter
df[(df['age']>18) & (df['country']=='US')]	Multi-condition
df.query("country=='US'")	SQL-style filter
df[df['age'].between(20,30)]	Range filter
df[df['id'].isin([1,2,3])]	Is-in filter

Cleaning & Transforming	
df.dropna(subset=['id'])	Drop missing
df.dropna(how='all')	Drop all-null rows
df.dropna(inplace=True)	Drop missing in-place
df.fillna(0)	Fill missing
df.fillna(0, inplace=True)	Fill in-place
df.ffill()	Forward fill
df.drop_duplicates(subset=['id'])	Deduplicate
df.duplicated()	Check duplicates
df.columns.difference(['id'])	Exclude columns
df.reset_index(drop=True)	Reset index
df.reset_index(drop=True, inplace=True)	Reset index in-place
df.rename(columns={'a':'A'})	Rename
df.rename(columns={'a':'A'}, inplace=True)	Rename in-place
df.drop(columns=['col'])	Drop column
df.drop(columns=['col'], inplace=True)	Drop column in-place
df.replace({'A':1})	Replace values
df.assign(new_col=df['x']*2)	Add column
np.where(df['x']>0,1,0)	Conditional column

Type Handling	
df['id'].astype('Int64')	Cast type
df['id'].astype(str)	To string
pd.to_datetime(df['ts'])	To datetime
df['ts'].dt.date	Extract date
df['ts'].dt.year	Extract year
df['ts'].dt.month	Extract month
df['ts'].dt.dayofweek	Day of week

Aggregation & Grouping	
`df.groupby('cat')['sales'].sum()`	Single col group
`df.groupby(['cat','region'])['sales'].sum()`	Multi-col group
`df.groupby(['cat','region']).agg({'a':'sum','b':'mean'})`	Multi-col + multi-agg
`df.groupby('cat').mean()`	Group mean
`df.groupby('id').agg({'a':'sum','b':'count'})`	Multi-agg
`df.groupby('cat').size()`	Group size
`df['id'].count()`	Count
`df['user_id'].nunique()`	Unique count
`df['country'].value_counts()`	Frequency
`df.sort_values('sales', ascending=False)`	Sort
`df.sort_values('sales', inplace=True)`	Sort in-place
`df.nlargest(5, 'sales')`	Top N rows

Joining & Combining	
`pd.merge(a, b, on='id', how='left')`	Left join
`pd.merge(a, b, on='id', how='right')`	Right join
`pd.merge(a, b, on='id', how='inner')`	Inner join
`pd.merge(a, b, on='id', how='outer')`	Outer join
`pd.merge(a, b, how='cross')`	Cross join
`df1.join(df2, how='left')`	Index join
`pd.concat([df1, df2])`	Stack DataFrames
`pd.concat([df1, df2], axis=1)`	Concat columns

Advanced Operations	
`df['ma7'] = df['sales'].rolling(7).mean()`	Rolling mean
`df['lag'] = df['x'].shift(1)`	Lag column
`df['r'] = df.groupby('cat')['sales'].rank()`	Rank
`df.groupby('cat').cumcount()`	Cumulative count per group
`df['x'].cumsum()`	Cumulative sum
`df.groupby('cat')['x'].cumsum()`	Cumsum per group
`df['x'].cummax()`	Cumulative max
`df['x'].cummin()`	Cumulative min
`df['x'].pct_change()`	Pct change
`df['x'].diff(1)`	Diff (lag diff)
`df.pivot_table(values='sales', index='cat', columns='region')`	Pivot
`df.melt(id_vars=['id'])`	Wide to long
`df.explode('col')`	Explode list col
`df['x2'] = df['x'].apply(lambda x: x*2)`	Apply
`df['email'].str.lower()`	String lower
`df['col'].str.contains('x')`	String contains
`df['col'].str.split(',')`	String split
`df.loc[df['age']<0, 'age'] = 0`	Update rows